A PROJECT TITLE

ON

A LEAKAGE-AWARE MULTI-VIEW MACHINE LEARNING FRAMEWORK FOR DETECTING FRAUDULENT ONLINE JOB POSTINGS

Project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

Nimma Lokesh Reddy	**2520030366**
Ramagiri Rishik Rao	**2520030333**

Under the Esteemed guidance of

Jyothsna Datti

Assistant Professor, CSE

CSE

Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

Name of Title: A Leakage-Aware Multi-View Machine Learning Framework for Detecting Fraudulent Online Job Postings

1. ABSTRACT

Online recruitment platforms make employment opportunities easier to access, but they can also be misused to publish fraudulent job advertisements that request money, collect personal information, or mislead applicants. Manual verification of a large number of postings may be slow and inconsistent; therefore, this project proposes a leakage-aware machine-learning and natural-language-processing framework that can classify a job advertisement as genuine or fraudulent. The project may use the Employment Scam Aegean Dataset (EMSCAD), distributed on Kaggle as "Real or Fake? Fake Job Posting Prediction" by Shivam Bansal. The dataset contains 17,880 job advertisements, including 17,014 genuine and 866 fraudulent records, with 18 fields such as title, location, company profile, description, requirements, benefits, and the target label fraudulent. To reduce unrealistically high performance caused by repeated templates, normalized duplicate job templates may be grouped before the training, validation, and test split. Text features can be represented through word-level and character-level TF-IDF, while behavioural and categorical attributes might provide additional evidence. Cost-sensitive Logistic Regression, calibrated Linear SVM, and selected tree-based baselines may be compared; SMOTE or ADASYN may be examined only on training data. At the end of the project, performance can be reported using the confusion matrix, accuracy, fraud-class precision, recall, F1-score, F2-score, ROC-AUC, and PR-AUC or average precision. Fraud recall and fraud F1-score may be treated as the main evaluation measures because missing a fraudulent posting can be more harmful than producing a small number of false alerts.

Dataset source: Kaggle - Real or Fake? Fake Job Posting Prediction

2. INTRODUCTION

The rapid growth of online recruitment has made job searching faster and more convenient, but it has also created opportunities for employment scams. Fraudulent advertisements may imitate genuine companies, promise unrealistic salaries, request registration fees, or collect sensitive applicant information. Because a single platform can receive thousands of postings, checking every advertisement manually may require considerable time and human effort. This motivates the development of an automated screening system that can support job seekers and recruitment platforms by identifying suspicious postings early and consistently.

Fake job-posting detection can be treated as an imbalanced binary text-classification problem in which 0 represents a genuine job and 1 represents a fraudulent job. The proposed study may use the EMSCAD dataset from Kaggle and can combine textual, behavioural, and categorical views of each advertisement. Word-level TF-IDF may capture important terms and phrases, character-level TF-IDF can identify spelling patterns and obfuscated words, and structured indicators might represent features such as telecommuting, company logo availability, screening questions, employment type, and required experience. A leakage-aware split may keep identical normalized templates within a single data partition, while feature extraction and any oversampling can be fitted only on the training data. Candidate models may be selected using validation results, and the untouched test set can be used once for final comparison. The project may finally present both overall and fraud-class metrics so that the model is judged by its ability to detect scams rather than by accuracy alone.

3. SOFTWARE AND HARDWARE REQUIREMENTS

The proposed project can be developed on a standard personal computer or a cloud notebook environment. Because TF-IDF and linear classifiers are generally CPU-friendly, a dedicated GPU may not be compulsory; however, additional memory and optional GPU access might reduce experimentation time for larger feature spaces or future deep-learning comparisons. The following software and hardware configuration may support dataset preparation, model development, evaluation, visualization, and documentation.

Category	Requirement	Suggested Specification	Purpose
Hardware	Processor	64-bit dual-core CPU or better	Data preparation and model training
Hardware	Memory	Minimum 8 GB RAM; 16 GB recommended	Supports large sparse TF-IDF feature matrices
Hardware	Storage	At least 5 GB free space	Stores dataset, notebooks, models, and results
Hardware	GPU	Optional	May help only for future deep-learning comparisons
Software	Operating system	Windows 10/11, Linux, or macOS	Provides the development environment
Software	Programming language	Python 3.10 or later	Implements preprocessing, training, and evaluation
Software	Development platform	Jupyter Notebook, Kaggle Notebook, or Google Colab	Supports reproducible experiments
Software	Core libraries	pandas, NumPy, scikit-learn, imbalanced-learn, SciPy	Processes data and builds machine-learning pipelines
Software	Visualization	Matplotlib and Seaborn	Creates confusion matrices and evaluation graphs
Data	Dataset	EMSCAD: Kaggle fake-job-posting dataset	Provides 17,880 labelled job advertisements

Signature of the Guide **Course Instructor**